

Multiple Choice (10 marks)

1. The lithosphere of a planet is the layer that consists of
 - (a) the softer rocky material of the mantle.
 - (b) the lava that comes out of volcanoes.
 - (c) material between the crust and the mantle.
 - (d) the rigid rocky material of the crust and uppermost portion of the mantle.
2. How do we know how old the Earth is?
 - (a) From the layering of materials within the Earth.
 - (b) From fossils of ancient life.
 - (c) From the cratering history of Earth's surface.
 - (d) From radioactive dating of rocks and meteorites.
3. If the Moon is setting at noon the phase of the Moon must be
 - (a) third quarter.
 - (b) waning crescent.
 - (c) waxing crescent.
 - (d) full.
4. You are pushing a truck along a road. Would it be easier to accelerate this truck on Mars? Why? (Assume there is no friction)
 - (a) It would be harder since the truck is heavier on Mars.
 - (b) It would be easier since the truck is lighter on Mars.
 - (c) It would be harder since the truck is lighter on Mars.
 - (d) It would be the same no matter where you are.
5. Which of the following lists the ingredients of the solar nebula from highest to lowest percentage of mass in the nebula?
 - (a) hydrogen compounds ($\text{H}_2\text{OCH}_4\text{NH}_3$) light gases (H He) metals rocks
 - (b) hydrogen compounds ($\text{H}_2\text{OCH}_4\text{NH}_3$) light gases (H He) rocks metals
 - (c) light gases (H He) hydrogen compounds ($\text{H}_2\text{OCH}_4\text{NH}_3$) metals rocks
 - (d) light gases (H He) hydrogen compounds ($\text{H}_2\text{OCH}_4\text{NH}_3$) rocks metals

True / False (10 marks)

Answer True or False

6. True or False: Planetary rings can form from the dismantling of small moons due to impacts from larger celestial bodies.
7. True or False: Venus shows no signs of volcanic activity, indicating a geologically inactive surface.
8. True or False: Barnard's Star is the second closest star system to Earth after Proxima Centauri.
9. True or False: The Sun's corona cannot be seen because it emits only infrared radiation, which is not visible to the naked eye.
10. True or False: In $20,000$ years, the Moon will be closer to the Earth and the length of the Earth's day will decrease due to gravitational effects.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of Betelgeuse within the constellation Orion, including its characteristics, role in stellar evolution, and its cultural impact in astronomy.
12. Discuss the implications of Kepler's laws of planetary motion and evaluate why the inverse square law of gravitational attraction is not a direct consequence of these laws.

End of Paper